

Grade 8
Unit 1
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

1. Convert $\frac{14}{9}$ into a decimal and a percent.

Decimal $1.\bar{5}$ Percent $155.\bar{5}\%$

2. Convert 0.12 into a fraction and a percent.

Fraction $\frac{3}{25}$ Percent 12%

$$\frac{12 \div 4}{100 \div 4} = \frac{3}{25}$$

3. Convert 45% into a decimal and fraction.

Decimal 0.45 Fraction $\frac{45}{100}$ or $\frac{9}{20}$

4. The fraction $\frac{5}{9}$ rewritten in decimal form is $0.\bar{5}$. This is a

- ☒ terminating decimal.
- ☐ repeating decimal.

5. The fraction $\frac{15}{8}$ rewritten in decimal form is 1.875 . This is a

- ☒ terminating decimal.
- ☐ repeating decimal.

6. Which of the following statements is true?

A. $\frac{2}{3} > 66\%$

B. $\frac{2}{3} < 66\%$

C. $\frac{2}{3} = 6.6\%$

D. $\frac{2}{3} = 66\%$

$$\frac{2}{3} = 0.\bar{6} = 66.\bar{6}\%$$

For 7-10, consider the repeating decimal $0.\bar{7}$.

7. If $x = 0.\bar{7}$, $10x = 7.\bar{7}$.

8. Explain why the subtraction of $10x$ and x results in the elimination of the repeating portion of the decimal.

$10x - x$ is equivalent to $7.\bar{7} - 0.\bar{7}$. Both values have $0.\bar{7}$ repeating, so subtracting the two values eliminates the values after the decimal.

$$7.\bar{7} - 0.\bar{7} = 7$$

9. Complete the statement.

$$10x - x = 7.\bar{7} - 0.\bar{7}$$

10. Solve the equation for x to determine the fraction equivalent in simplest form.

$$9x = 7$$

$$x = \frac{7}{9}$$